# Trevor Chen

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## Education

**California Institute of Technology** — Pasadena, CA

*B.S. in Applied and Computational Mathematics | GPA: 4.0/4.0* — *Expected Jun. 2029*

- **Relevant Coursework:** Agentic Systems, Diffusion Models, Generative Models, Deep Learning, ML & Data Mining, GPU Programming (CUDA), Operating Systems, Statistical Inference, Probability Models, Applied Linear Algebra
- **Honors:** Y Combinator F25 Acceptance; YC AI Startup School; USACO Platinum; Regeneron STS Scholar; SPIE Scholarship; AIME Qualifier; National Merit Scholar; International Logic Olympiad Silver; Outstanding Presentation Award, Dayton-Cincinnati Aerospace Sciences Conference

## Experience

**Ascade (Accepted to Y Combinator F25)** — Jun. 2025 – May 2026

*Founder & CTO* — *San Diego, CA*

- Built and shipped a formally verified payment processor to production; it reached **$25K MRR**
- Won a **DARPA ERIS** contract and other federal work to build formally verified access control (IAM) for AI agents
- Trained an 8B model that beats frontier models, including Claude Opus 4.6, on formal-verification benchmarks

**Leonardo DRS Daylight Solutions** — Jun. 2025 – Aug. 2025

*Machine Learning Engineer* — *San Diego, CA*

- Built custom software for Daylight's laser research team, including user interfaces and automated real-time analysis of sensor outputs
- Built a filtering model with a custom cost function that cut water-vapor noise in laser spectroscopy data by **98%+**
- Replaced the cost function with a piecewise approximation, making the software **50%** faster than the legacy version
- Shipped the model in Daylight's MIRcat laser software, improving signal-to-noise by **99%** in humid conditions

**Spectral Energies** — Feb. 2022 – Jul. 2025

*R&D Intern* — *Remote*

- Designed broadband mid-IR diffraction gratings for ultrafast lasers with **>90%** efficiency over 3.6–6.6 µm; prototypes measured **92%** against 94.2% theoretical
- Co-developed the fabrication process with Penn State Nanofab (electron-beam lithography and anisotropic wet etching) for smooth blazed grooves, a **30%** efficiency gain; optimized Au–$Al_2O_3$ coatings for efficiency and laser damage threshold
- Found a "double-blazing" groove design in simulation (**+50%** bandwidth, **+20%** efficiency); simulation pipelines cut prototype cycles by **70%** and manufacturing error by **80%**
- Delivered 10 gratings to 5+ national labs, supporting **$1.3M+** in R&D contracts

## Research Experience

**ETH Zurich AI Center, Krause Group** (joint with Arnold Lab, Caltech) — Oct. 2026 – Present

*Researcher, with Prof. Andreas Krause and Prof. Frances Arnold* — *Remote*

- Building a closed-loop discovery system for enzyme and protein design: an LLM proposes mechanistic hypotheses, each is distilled into a flow model that generates designs, and wet-lab results confirm or refute it
- Developing signed distillation, which folds confirmed hypotheses into the generator and suppresses refuted ones while staying anchored to the previous model, for campaigns of only $\sim 10^2$ wet-lab measurements
- Comparing against active search, reward-based fine-tuning and agentic LLM loops, with wet-lab validation in Frances Arnold's lab

**Yue Lab, Caltech** — Jan. 2026 – Present

*Undergraduate Researcher, advised by Prof. Yisong Yue* — *Pasadena, CA*

- Led an 11-author study with Genesis Molecular AI on steering diffusion models for drug discovery with reinforcement learning and active learning; it reaches the best baseline's score with **58**× fewer evaluations (**NeurIPS 2026**)
- Showed that extra latent dimensions delay memorization in diffusion models (**3.6**× later on CelebA) and explained it through training dynamics (**ICML 2026 HiLD**; **ICLR 2027** under review)
- Ran distributed training on multi-node GPU clusters (Slurm) with fault-tolerant scheduling

**Caltech SURF Fellow, Perona Vision Lab** — Jun. 2026 – Aug. 2026

*Summer Undergraduate Research Fellow, advised by Prof. Pietro Perona* — *Pasadena, CA*

- Found a family of training objectives for sparse autoencoders (SAEs) that choose which features the SAE learns, so the dictionary can be picked on purpose instead of left to the random seed (**CVPR 2027**, under review)
- Showed with linear probes and planted-signal recovery that dictionaries from different objectives differ in what they encode and recover
- Trained a diffusion transformer from scratch and **36** SAEs (**220K+** features) on every layer's residual stream and attention/MLP outputs, then ablated features across denoising steps to trace how they drive generation

**UCSD San Diego Supercomputer Center** Oct. 2020 – May 2025
*Computational Biology Research Intern* *San Diego, CA*

- Built ML models that predict protein–ligand binding at **87%** accuracy and screened **50K+** compounds on HPC clusters for HIV and diabetic-cataract drugs
- Raised binding-prediction accuracy by **20%** with docking-simulation data augmentation (3 papers)
- Mentored 3 interns, running docking simulations to validate their ML models

## Projects

**Breaking Pangram's AI-Image Detector** | *Independent Red-Teaming* Aug. 2026

- Red-teamed a commercial AI-image detector in both directions: adversarially trained autoencoders (VAE, GAN) make arbitrary AI images pass as real, and ordinary upscaling gets real photos flagged as AI
- Disclosed the findings to Pangram's co-founder and withheld the attack code to prevent misuse

**Real-Time CUDA Black Hole Renderer** | *CS 179 GPU Programming* Jun. 2026

- Co-built a CUDA/C++ ray tracer that bends light through curved spacetime per pixel, rendering gravitational lensing, Einstein rings and a Doppler-shifted accretion disk in real time
- Wrote the CUDA rendering and post-processing kernels (bloom, HDR tone mapping) and a browser viewer with live controls and FPS profiling
- Validated against general relativity: light deflection within **0.44%** of theory, energy drift $<\mathbf{10}^{-6}$, a 12-test physics suite and CPU–GPU parity tests

**Trading on Binary Prediction Markets** | *Independent Project* May 2025 – Aug. 2025

- Built an algorithmic trading system with a reinforcement-learning inventory manager (**30%** less slippage) and sentiment signals
- Arbitrage and hedging strategies returned about **8%** a week over 3 months, on an execution API **6**× faster than the first version

## Skills

**Languages**: Python, C++, C, CUDA, SQL, Lean 4, Dafny
**ML**: PyTorch, Transformers, Diffusion Models, Reinforcement Learning, LLM Fine-Tuning, Active Learning, Evals
**Interpretability & Safety**: Sparse Autoencoders, Causal Interventions, Linear Probing, Red-Teaming
**Infrastructure**: Distributed Training, Multi-GPU, HPC, Slurm, Linux, Git
**AI for Science**: Boltz, ESM3, GenMol, DISCO, Protein Engineering, Molecular Optimization
**Other**: Formal Verification, AI Agents, Nanofabrication, Optical Simulation

## Publications & Invited Talks

[*] *Equal contribution (shared first author)*

**T. Chen**[*], A. Dai[*], J. Yang, R. De Santi, *et al.*, M. Al-Shedivat, Y. Yue. *On the Design Space of Discrete Diffusion Online Adaptation for Molecular Optimization*. **NeurIPS 2026**.

**T. Chen**, R. Shahbaba. *How Excess Latent Dimensionality Delays Memorization in Diffusion Models*. **ICLR 2027** (under review).

**T. Chen**, R. Shahbaba, K. Peng, A. Garg. *How Excess Latent Dimensionality Delays Memorization in Diffusion Models*. **ICML 2026 Workshop on High-dimensional Learning Dynamics (HiLD)**.

**T. Chen**, I. Demler, P. Perona. *Objective Geometry Steers Sparse Autoencoder Vocabularies*. **CVPR 2027** (under review).

J. Y. Tang[*], **T. Chen**[*], V. L. Kouznetsova, I. F. Tsigelny. *Anelloviruses and Cancer*. **The Journal of Infectious Diseases** 231(2), 2025.

**T. Chen**, B. Liu, J. A. McCoy, *et al. Development of Broadband High-Efficiency Mid-IR Gratings for High-Energy Ultrafast Lasers*. **Optical Materials Express** 14(5), 2024.

**T. Chen**, P. S. Hsu. *Enabling Ultra-Broadband High-Efficiency Gratings for Single-Cycle Infrared Lasers*. **SPIE Nanoengineering**, 2024.

P. S. Hsu[*], **T. Chen**[*], A. Roy, *et al. A Compact LWIR Borescope Sensor for 2D Engine Component Surface Temperature Measurement*. **Measurement Science and Technology** 34(6), 2023.

A. Roy, P. S. Hsu, A. Zhang, **T. Chen**, *et al. Development of Compact LWIR Borescope Sensor for Non-contact 2D Engine Component Surface Temperature Measurement*. **AIAA AVIATION Forum**, 2023.

**T. Chen**, A. Zhang, A. Roy, P. S. Hsu, O. Bibik, S. Adhikari. *2D High Temperature Sensor Development Using LWIR Borescope*. **SPIE Infrared Sensors, Devices, and Applications XIII**, 2023.

**T. Chen**, R. Chen, A. You, V. L. Kouznetsova, I. F. Tsigelny. *Search of Inhibitors of Aldose Reductase for Treatment of Diabetic Cataracts Using Machine Learning*. **Advances in Ophthalmology Practice and Research** 3(4), 2023.

A. K. Gao, **T. Chen**, V. L. Kouznetsova, I. F. Tsigelny. *Machine-Learning-Based Virtual Screening and Ligand Docking Identify Potent HIV-1 Protease Inhibitors*. **Artificial Intelligence Chemistry** 1(2), 2023.

**T. Chen**, P. S. Hsu. *Compact Borescope Design for 2D Long-Wavelength IR Thermal Imaging Sensing*. **Optica LACSEA**, 2022.

*Invited Talks*

**Invited Ethics Speaker**, iGEM Conference.
**Invited Researcher**, Leonardo DRS Daylight Solutions Executive Team.